

RESEARCH ARTICLE

A Deep Learning-Based Ultra-Lightweight Architecture for Atrial Fibrillation Detection Using Single-Lead ECG Recordings

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ABSTRACT Atrial fibrillation (A-Fib), a common cardiac arrhythmia, is linked to serious complications like heart failure and stroke, which early diagnosis could prevent. However, its paroxysmal and often asymptomatic nature complicates detection. In this paper, we propose a novel deep-learning model for single-lead ECG signal analysis, combining a convolutional neural network (CNN) to extract localized features and a long short-term memory (LSTM) network to capture temporal dependencies. To address data imbalance in the PhysioNet/CinC Challenge 2017 dataset, we employ a filtering-based approach with Neurokit, BioSPPy, and EngZeeMod for synthetic data generation. Simulation results show the model achieves a 99.56% F1 score, a 102.25 dB gain in efficiency, a 1.06% performance gain over the state-of-the-art method, and real-time diagnosis with a 0.48×10^{-3} second inference time. These simulation results confirm the superior performance of the proposed model.

INDEX TERMS Atrial fibrillation, ECG, ultra-lightweight, hybrid, real-time.

I. INTRODUCTION

The upper and lower chambers of the heart contract erratically and asynchronously in atrial fibrillation (A-Fib) [1], [2], [3]. This irregular rhythm is strongly associated with various cardiac complications, such as stroke, transient ischemic attack (TIA), and heart failure, and can be caused by various lifestyle and environmental factors, including excessive alcohol consumption, drugs, smoking, stress, and overexertion [4], [5], [6], [7]. In the United States only, approximately 2.3 million adults are affected by A-Fib [8]. It is predicted that by the year 2050, approximately 6-12 million individuals in the United States and around 18 million in Europe by 2060 will be affected by A-Fib [9], [10], [11], [12]. It is pertinent to mention that the A-Fib is a permanent

condition [13]. Early detection of atrial fibrillation is essential for timely intervention and management, allowing healthcare professionals to implement appropriate treatments such as anticoagulation therapy to reduce the risk of stroke.

ECG signals have different waves and parts that show different stages of the heart's cycle, as shown in Fig. 1. The P wave represents the contraction of the atria, the QRS complex indicates the contraction of the ventricles, the T wave signifies the relaxation of the ventricles, and the ST segment denotes the interval between the end of ventricular contraction and the beginning of relaxation. Analyzing ECG signals can deliver information about the heart rate, rhythm, and electrical behavior of the heart [14]. This helps doctors to decide on treatment options, which might include medications, electrical cardioversion (a controlled shock to the heart), catheter ablation (a procedure to fix heart rhythm problems), or surgery. However, understanding ECG readings

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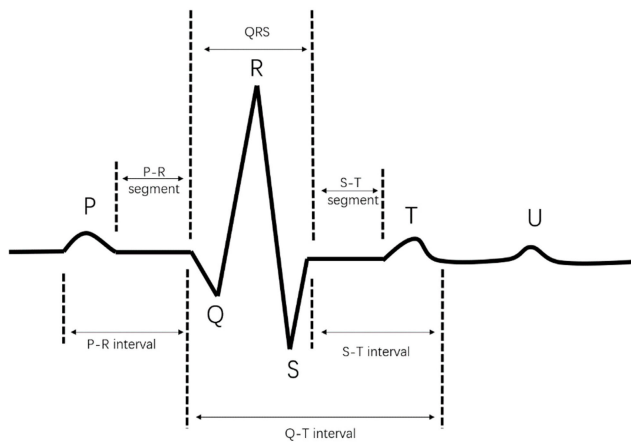


FIGURE 1. Segments of an ECG signal.

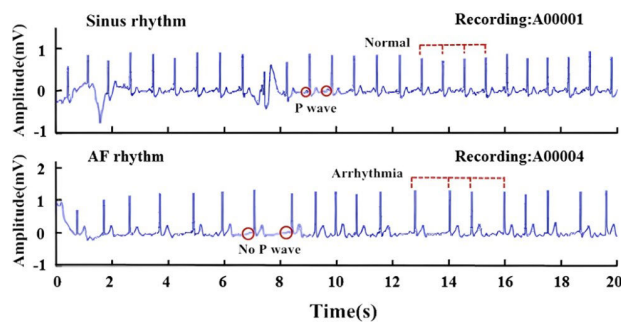


FIGURE 2. Sinus rhythm versus AF rhythm.

requires special training and accurate equipment, which may not always be available [15].

An A-Fib rhythm differs from a normal rhythm in both temporal fluctuations, such as irregular arrhythmic patterns, and structural characteristics, such as the absence of a P wave. Figure 2 presents a digitized ECG signal example from ECG records, a healthy individual's heart exhibits a regular contraction and relaxation pattern, maintaining a sinus rhythm of 60 to 100 beats per minute. In contrast, with an A-Fib rhythmic activity, the heart's electrical behavior becomes disorderly and accelerated, exceeding the typical sinus rhythm rate.

Given the large population affected by A-Fib and its sporadic occurrence, it necessitates an easily accessible diagnostic mechanism. Advancements in technology have enhanced the availability of portable and wearable ECG equipment for detecting A-Fib [16]. Compared to the 12-lead ECG, the portable single-lead ECG devices offer an instant, easy, and location-independent approach for heart rhythm monitoring. These devices are potentially beneficial in continuous and real-time monitoring of ECG signals in an outpatient ambulatory setting, such as at home or work. Despite having several advantages, these gadgets have certain disadvantages as well, which include a small capturing window, a vulnerability to wearer-induced artifacts, and

degraded signal quality. As a result, it's critical to create novel algorithms that can accurately identify A-Fib from single-lead ECG readings and take into consideration the technological difficulties these devices provide.

Various methods for A-Fib detection have been employed to automate the interpretation of ECG signals, containing both traditional machine learning and deep learning approaches [17]. Traditional machine learning methods rely on extracting hand-crafted features that capture the specific attributes of A-Fib from the ECG data. These extracted features are then utilized as inputs for established classifiers like K-nearest neighbors (KNNs) [18], Random Forests [19], linear discriminant analysis [20], support vector machine (SVM) [21], AdaBoost [22], and handcrafted features capturing ECG characteristics combined with DWT, CWT, and XGBoost [23] are used to perform the recognition of A-Fib patterns.

Deep learning has emerged as a powerful tool for data analysis across diverse fields, outperforming traditional methods in ECG signal classification, especially with large datasets. It bypasses manual feature extraction, delivering accurate results. CNNs excel in autonomously learning critical features from raw ECG data, capturing localized patterns vital for classification [24]. Meanwhile, LSTM networks, a type of RNN, effectively model temporal dynamics in time series data like ECGs, leveraging their memory structure to retain relevant information [25], [26]. This efficacy extends beyond ECGs to areas like image processing [24], audio analysis [25], and speech recognition [27].

The key objectives achieved in this paper are listed below:

- To propose a cost-effective and ultra-lightweight hybrid CNN-LSTM model for real-time A-Fib detection using single-lead ECG recordings.
- To utilize the raw ECG signal as input for the proposed CNN-LSTM model for automatic A-Fib detection, eliminating time-consuming preprocessing steps.
- To propose a reliable data augmentation method to address the challenge of limited and imbalanced ECG data.

In summary, the goal of this work is to provide a computationally cost-effective tool for analyzing ECG signals with higher reliability.

This paper is structured as follows: Section II surveys prior research relevant to our work. Section III presents and details the proposed model. Section IV assesses its performance. Section V examines how different parameters influence the model's effectiveness. Section VI concludes the paper.

II. RELATED WORK

The AF-GCN algorithm introduces a new way to detect atrial fibrillation using graph convolution networks (GCN). This technique effectively combines features of ECG shape and rhythm, overcoming limitations seen in other methods. Their results show that AF-GCN performs better, making it a potential algorithm for accurate A-Fib detection in wearable devices [28].

A lightweight Atrial Fibrillation detection based on an improved Variable Step Dynamic Threshold Local Binary Pattern algorithm is proposed in [29]. The authors in [29] investigate the ability of Videoplethysmography (VPG) based on high-end Basler RGB camera to identify AF in a controlled hospital setting.

The CWT-MB-ResNet framework successfully identifies A-Fib using single-lead ECG, addressing data imbalances, and showing promise for improving heart disease diagnosis [30]. In [31], a real-time classification model for A-Fib, utilizing RR-Intervals obtained from ECG data, whereby SVM models incorporating extracted attributes achieve moderate accuracy, yet challenges arise from differences across databases, hindering the desired sensitivity in the 2017 Challenge dataset. To deal with the lack of labeled A-Fib data, the AFE-GAN model suggests generating A-Fib-like ECG signals from normal ones. The model achieves an accuracy of 72.4% for A-Fib detection [32]. The PFSV-AGAN model [33] utilizes ECG signals for categorizing arrhythmias and handles varying data sizes through adversarial neural networks. This model identifies abnormalities, surpassing traditional methods, emphasizing the importance of feature selection and parameter tuning in addressing arrhythmia disorders. Many studies explore automated classification techniques, using methods like wavelet [34], filter banks [35], support vector machines [36], and other machine learning methods [37], [38] for detecting heart arrhythmias through ECG signals. The work done in [39] uncovers complex non-linear correlations in identifying cardiac illnesses from ECG analysis.

In a deep learning model, a series of processing layers work simultaneously to understand data features through multiple stages of abstraction [40], [41], [42]. A 1D-CNN architecture [43] is proposed to classify various forms of cardiac arrhythmias. These developments demonstrate the versatility of CNNs in managing diverse data modalities beyond images [44]. The supreme CNN architecture (SCA) achieves 99.17% accuracy for A-Fib classification while the software-selected CNN Architecture (SSCA) provides 98.95% accuracy [45].

Long short-term memory (LSTM) is an advanced form of the recurrent neural network (RNN) architecture, commonly utilized in deep learning for handling sequential input. In [46], an LSTM-based network is combined with automatic R-peak recognition via convolutional networks for ECG data. An automated method for A-Fib detection is suggested using an LSTM network on the MIT-BIH dataset, getting an accuracy of 98.51% [47]. For detecting myocardial infarction from 12-lead ECG data, a hybrid RNN (LSTM) method gets an average accuracy of 93.08% [48]. BiLSTM and CLSTM networks are tested in conjunction with spectral feature-based classification, giving a training accuracy of 97.88%. Additionally, P-wave detection displays the best test accuracy of 87.65%, supporting BiLSTM's effectiveness in capturing time-sensitive ECG features [49]. In [50],

an LSTM layer and two fully linked layers are used to categorize beats from the MIT-BIH Arrhythmia Database into eight groups. To differentiate five distinct types of ECG beats, the same data is additionally supplied into two completely linked layers and two bidirectional LSTM layers [51]. A further study examines the efficacy of six different LSTM network topologies for ECG classification, using the MIT-BIH Arrhythmia and Normal Sinus Rhythm Databases [52].

The authors in [53] introduce CKConv algorithm built on LSTM architecture, aimed at enhancing the prediction of A-Fib detection using data from general practitioners. This CKConv algorithm is designed to manage irregularities in medical data and handle extended sequences of temporal information. By incorporating temporal data for outcome prediction, it demonstrates superior performance compared to non-temporal algorithms.

III. METHODOLOGY

In this section, we will explain the dataset, data augmentation strategy, modeling philosophy, architecture, hyper-parameter details, evaluation metrics, and the loss function.

A. PHYSIONET/CINC CHALLENGE DATASET

The A-Fib classification dataset provided by the PhysioNet/CinC Challenge 2017 is used for the investigation [54]. This dataset comprises single-lead ECG recordings from diverse sources, classified into four categories: normal sinus rhythm, A-Fib, other rhythms, and noisy recordings. The ECG recordings were obtained using the AliveCor device, band-pass filtered, with durations ranging from 9 to 61 seconds, and sampled at 300 Hz. Expert cardiologists manually assigned labels based on the dominant rhythm in each recording. The dataset includes 8,528 publicly available training recordings. To examine binary classification, we focused on the normal sinus rhythm and A-Fib classes, which contain 5154 and 771 recordings, respectively. The normal class does not include any non-A-Fib abnormalities, resulting in a highly imbalanced dataset where normal class recordings account for 87% of the total dataset.

B. DATA AUGMENTATION STRATEGY

Data augmentation is used to enhance the quality and quantity of the dataset, addressing challenges such as class imbalance and the potential of model overfitting. Given the difficulty in acquiring diverse ECG data from a large number of A-Fib patients in real-world healthcare settings, the available dataset primarily comprises recordings from individuals without A-Fib, leading to limited instances of authentic A-Fib data. To mitigate these issues, a data augmentation technique using multiple ECG filters is used and combined with the outputs as explained in the next section.

C. PROPOSED NBE FILTERS FOR DATA AUGMENTATION

We proposed a data augmentation technique using three distinct ECG filters: Neurokit [55], BioSPPy [50] and

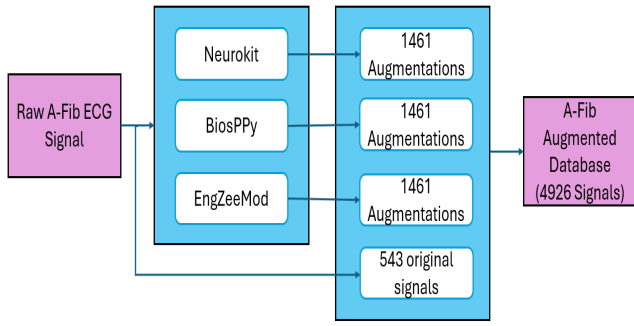


FIGURE 3. Block diagram of AFIB signal's database generation.

EngZeeMod [56] all together referred as NBE filters for the sake of simplicity, as shown in Fig. 3, to address class imbalance in the PhysioNet/CinC Challenge 2017 dataset.

1) NEUROKIT FILTER

The Neurokit filtering method utilizes a fifth-order high-pass Butterworth filter to remove slow drift and DC offset [55]. The Butterworth high pass filter in eq. (1) with ($n=5$) is used with a low cutoff frequency f_c of 0.5 Hz to process the input frequency f . Moreover, the Neurokit method eliminates 50 Hz power line noise through signal smoothing with a 50 Hz period moving average kernel.

$$H(f) = \frac{1}{\sqrt{1 + \left(\frac{f_c}{f}\right)^{2n}}} \quad (1)$$

2) BIOSPPY FILTER

BioSPPy is used in the second filtering approach proposed by Carreiras et al. [57]. The finite impulse response (FIR) bandpass filter removes ECG signal frequencies below 3 Hz and above 45 Hz. By processing the input signal twice, once forward and once backward, the method makes use of a zero-phase linear digital filter. To ensure an odd order, the order of the filter, represented by N , is calculated as 0.3 times the sampling rate. If the result is even, one is added. Before calculating the FIR filter using eq. (2), the frequency is first normalized to the Nyquist frequency. The input signal at each tap, $x(n-i)$, corresponds to the filter order. The filter coefficients are denoted by b_i within the range $0 \leq i \leq N$. The resulting filtered output signal is denoted as $y(n)$.

$$y(n) = \sum_{i=0}^N b_i x(n-i) \quad (2)$$

3) ENGZEEMOD FILTER

Engelse and C. Zeelenberg, in eq. (3), presented a fourth-order Butterworth band-stop filter. This filter is designed to efficiently attenuate frequencies within a defined range while permitting frequencies outside this range to pass with minimal interference.

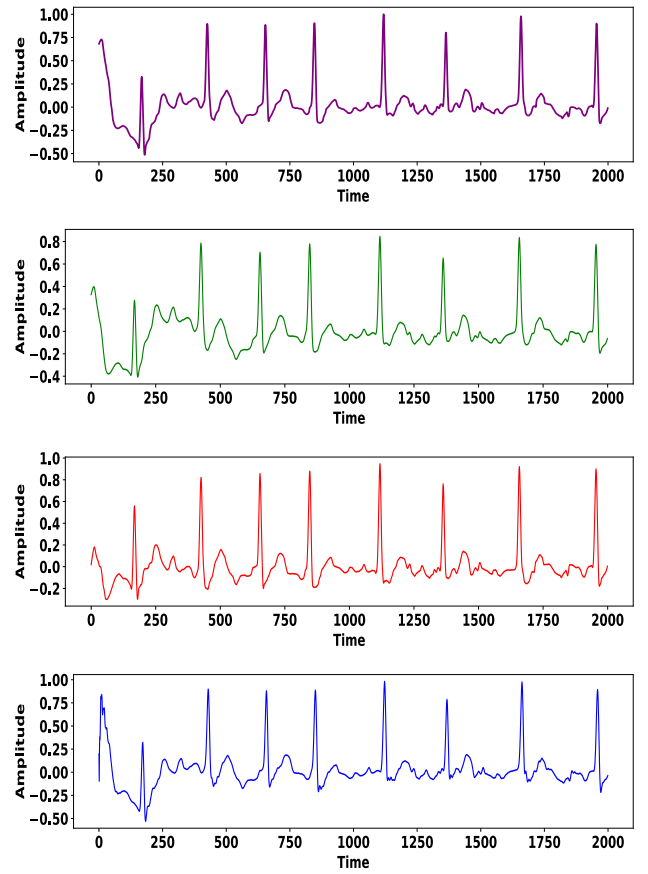


FIGURE 4. a) Raw ECG and filtered signals using b) Neurokit c) BiosPPy d) EngZeeMod.

This filter is characterized by cutoff frequencies f_{cl} and f_{ch} set at 30 Hz and 60 Hz, respectively. Frequencies between f_{cl} and f_{ch} are suppressed, making it ideal for applications where specific frequency components need to be eliminated or reduced, such as in noise cancellation or interference mitigation [58].

$$B(f) = \frac{1}{\sqrt{1 + \left(\frac{f}{f_{ch}}\right)^{2n}}} \cdot \frac{1}{\sqrt{1 + \left(\frac{f_{cl}}{f}\right)^{2n}}} \quad (3)$$

The augmentation technique demonstrates that multi-filtering produces smoother ECG signals with subtle pattern variations. Using multiple filters with different functionalities introduces a wider range of transformations to resolve the class imbalance by providing sufficient A-Fib class signals.

Figure 4, the collective outcomes of the A-Fib signals are visually depicted, encompassing the outcomes produced by all three filters. The signals generated by these filters are visually differentiated by colors as shown in these subfigures. The filters exhibited a very good resemblance to the original ECG signal which is quantified using the Cosine Similarity (CS) Index explained in next sub-section.

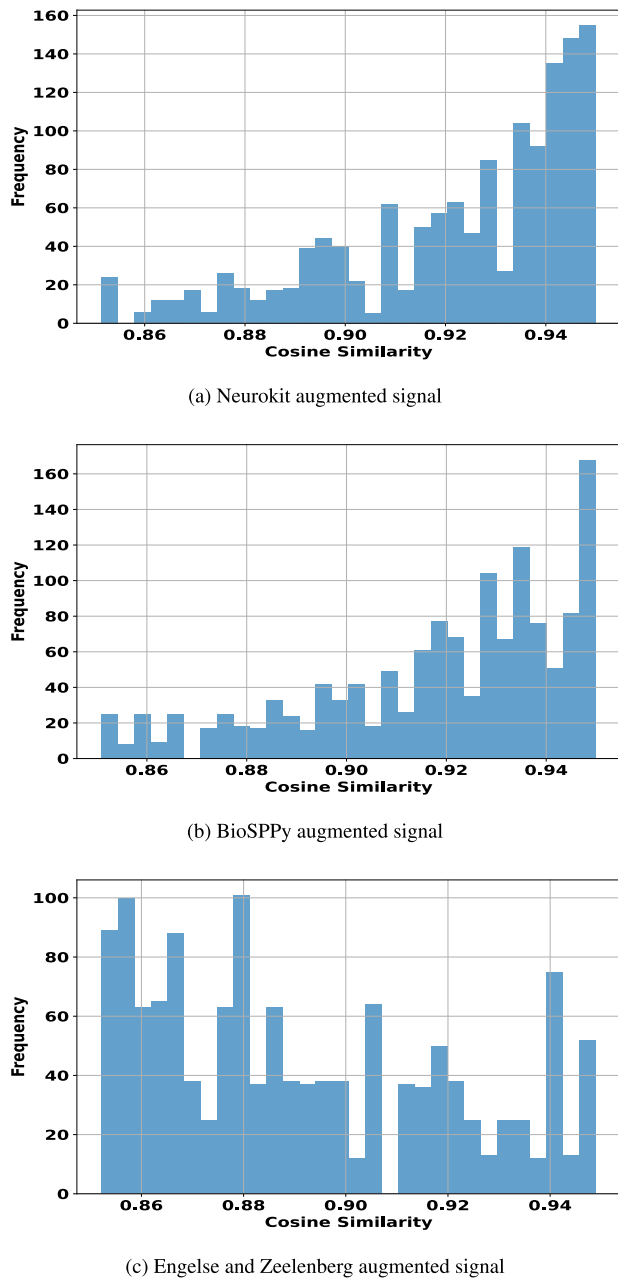


FIGURE 5. Similarity index comparisons of Raw ECG signal with various augmented signals.

4) COSINE SIMILARITY INDEX

Figure 4 depicts that the NBE filters produce signals having a high correlation with the input signal. However, this is not always true and a careful examination is required before adding the newly produced signal in the database. The augmented signal should not be a replica of the input signal nor there should be too much variation between them rendering the sample useless.

Cosine similarity, measures the similarity between ECG signals, provides an efficient and robust method for comparing and classifying signals in terms of matched features

compared with the input signal given by eq. (4):

$$CS(A, B) = \frac{A \cdot B}{\|A\| \cdot \|B\|} \quad (4)$$

Here, A and B are vectors representing the raw signal and newly generated instances of A-Fib data. Among the samples generated using Neurokit and BioSPPy filters (refer to Fig. 5), those exhibiting a higher similarity index ranging from $\delta_1 = 0.85$ to $\delta_2 = 0.95$, displaying almost identical features as compared to the original raw A-Fib sample yet subtly different, were added to the database and the rest of the samples were discarded to avoid replication of the input raw A-Fib signal. A brief description of data generation is given by Algorithm 1.

Algorithm 1 Data Augmentation Strategy

```

1: Input: Raw AF ECG signals
2: Output: Augmented database
3: for  $i = 1$  to  $N_f$  do  $\triangleright N_f$ : Number of filters
4:   for  $s = 1$  to  $L$  do  $\triangleright L$ : Maximum number of raw samples
5:      $j \leftarrow \text{AugmentedSignal}(i, s)$   $\triangleright$  Generate augmented signal
6:     Compute cosine similarity  $CS(i, j)$  using Eq. (4)
7:     if  $CS(i, j) > \delta_1$  and  $CS(i, j) < \delta_2$  then
8:       Add augmented signal  $j$  to database
9:     else
10:      Discard  $j$ 
11:    end if
12:  end for
13: end for

```

D. DESIGN PHILOSOPHY

The design principle of the proposed model, as shown in Fig. 6, is based on hybrid architecture [14], [26], [59]. First, the raw ECG signals are used to bypass laborious pre-processing, preserving essential information and enabling the model to capture inherent patterns more effectively. Secondly, our hybrid architecture, integrating CNNs and LSTM, allows for the simultaneous extraction of spatial and temporal features. It is important to comprehensively understand the spatial characteristics of ECG waveforms and the temporal dependencies over time for the detection of A-Fib. These combined strategies underscore the model's robustness and efficacy in the intricate task of detecting A-Fib from single-lead ECG signals.

The following subsections provide further details about our model in terms of the model architecture, hyper-parameters, and the experimental setup.

E. MODEL HYPER-PARAMETERS AND EXPERIMENTAL SETUP

1) MODEL

The architecture of proposed hybrid model represented by Fig. 7, consisting of CNN and LSTM layers. First, a 1D CNN

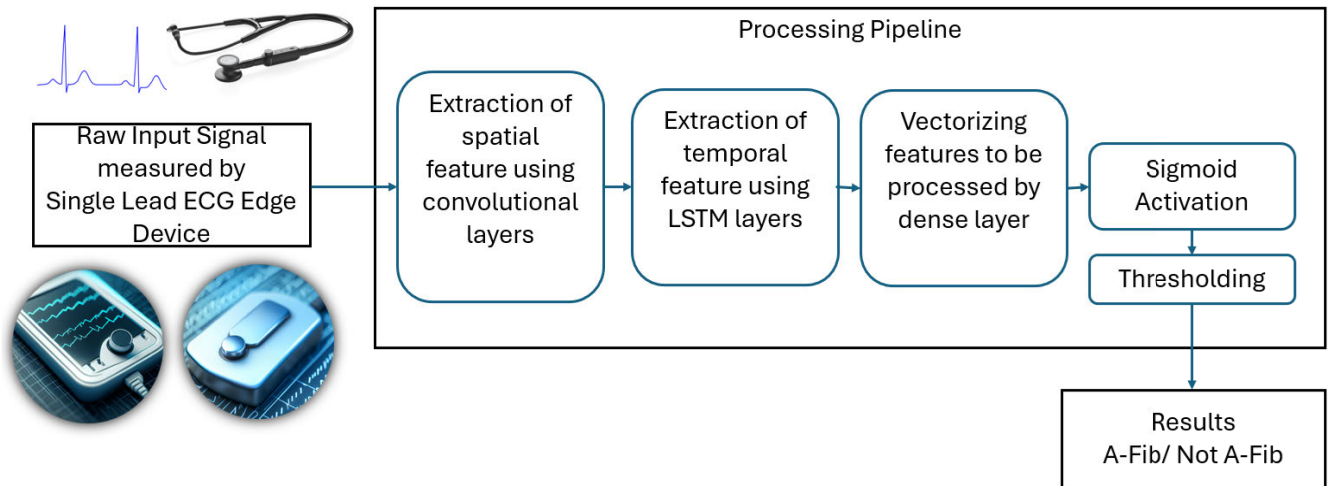


FIGURE 6. The system level representation of hybrid CNN-LSTM architecture.

layer as a feature extractor with only 16 units to extract the most important features from short raw ECG signals. These features are then extrapolated using another 1D CNN with 36 units. These features, extracted by 1D CNN are providing information about the most significant features required for analyzing A-Fib/sinus signals. However, these are the local features with respect to the given kernel. To understand the relationship between these features, a global feature extractor is required, which in our case, is LSTM. The LSTM layers learn the global features required for analyzing the temporal information. Finally, these temporal features are processed by the fully connected layers, and a binary output is produced.

2) HYPER-PARAMETERS

The model employs two 1D convolutional layers with 16 and 36 filters, respectively, each using a kernel size of 5 and a stride of 2. The kernel size for the max-pooling layers is 5, while the stride is set to 3 and 1, respectively. These are followed by 2 LSTM layer having 64 hidden units. Then, the fully connected layers are deployed, having 36 and 16 neurons each. Finally, the output layer is deployed having 1 neuron to cater the binary decision having sigmoid as the activation function.

The model utilizes the Adam optimizer with a batch size of 42 and a learning rate set to 0.001. It was trained for 200 epochs, employing accuracy as the performance metric and binary cross-entropy as the loss function. Dropout and batch normalization approaches were used to avoid overfitting and increase generalization. Specifically, for LSTM layers a dropout = 0.2 is applied for both layers. For the fully connected layers, followed by LSTM layers, a dropout = 0.5 is applied. Moreover, L_2 regularization is applied to the penultimate fully connected layer. The details are summarized in Table 1.

TABLE 1. Model summary of the proposed CNN-LSTM network.

Layer (Type)	Output Shape	Parameters
conv1d_2 (Conv1D)	(None, 998, 16)	96
max_pooling1d_2 (MaxPooling1D)	(None, 332, 16)	0
conv1d_3 (Conv1D)	(None, 164, 36)	2916
max_pooling1d_3 (MaxPooling1D)	(None, 160, 36)	0
batch_normalization_1	(None, 160, 36)	144
lstm_1 (LSTM)	(None, 160, 64)	25856
lstm_2 (LSTM)	(None, 64)	33024
dense_1 (Dense)	(None, 36)	2340
drouput_1 (Dropout)	(None, 36)	0
dense_2 (Dense)	(None, 16)	592
drouput_2 (Dropout)	(None, 16)	0
dense_4 (Dense)	(None, 1)	17
Total Params: 64985		
Trainable Params: 64913		
Non-trainable Params: 72		

TABLE 2. Details of experimental setup.

Parameters	Value
Epochs	200
Learning rate	0.001
Loss function	Binary cross-entropy
Optimizer	Adam
Callbacks	Early stopping
Restore best weights	True

3) EXPERIMENTAL SETUP

The following setup is used to train the model as mentioned in Table 2.

The dataset was first divided into training, validation, and test sets. The validation set consists of 114 A-Fib signals, approximately 15% of the A-Fib data and 114 sinus signals (to keep the sample ratio same). The same distribution is followed to create the test data. Then, by following Algorithm 1, 4383 recordings were selected for the A-Fib

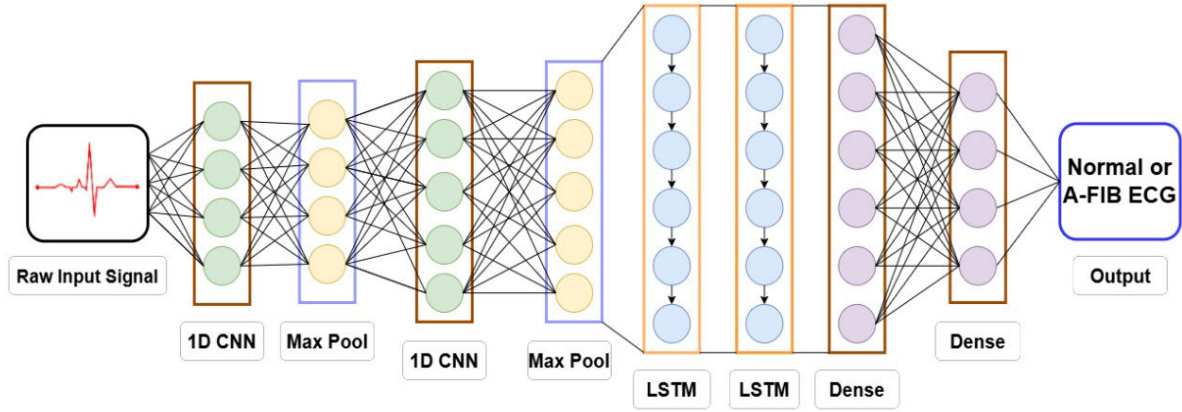


FIGURE 7. The architecture of the proposed hybrid CNN-LSTM model.

class. The resulting dataset contains 4926 recordings each for the normal and A-Fib class respectively.

Algorithm 2 Algorithm for ECG Signal Analysis

- 1: **Input:** Raw ECG signal
 - 2: **Output:** Classification of ECG signal
 - 3: **Training Phase:**
 - 4: Generate training data using Algorithm 1
 - 5: Train the network until convergence
 - 6: **Inference Phase:**
 - 7: Acquire the signal to be classified
 - 8: Feed acquired signal to the proposed network
 - 9: Network extracts features and performs classification
 - 10: **Return:** Classification results
-

F. EVALUATION METRICS

We choose classification accuracy, precision, recall, and F1-score performance measures for the evaluation of methodological tasks. These metrics offer valuable perspectives on the model's performance from various perspectives. The classification accuracy (ACC), represented in eq. (5) quantifies the proportion of accurately predicted instances for individual classes within the dataset.

$$Acc = \frac{\sum_{i=1}^n CM(i, i)}{\sum_{i=1}^n \sum_{j=1}^n CM(i, j)} \quad (5)$$

The Precision ($Pre(i)$), indicated eq. (6) indicates what percentage of identifications were correctly assigned to class i based on the total number of samples in that class.

$$Pre(i) = \frac{CM(i, i)}{\sum_{m=1}^n CM(m, i)} \quad (6)$$

where the $\sum_{m=1}^n CM(m, i)$ demonstrates the total number of samples that were classified to class i .

Recall $Rec(i)$, also known as the true positive rate (TPR) or sensitivity, is the predictor's capacity to locate all pertinent testing samples inside the dataset and is given in eq. (7).

$$Rec(i) = \frac{CM(i, i)}{\sum_{m=1}^n CM(i, m)} \quad (7)$$

where $\sum_{m=1}^n CM(i, m)$ indicates the total number of testing samples that are classified as belonging to class i . In this equation, n represents the total number of testing samples, and m represents the total number of classes.

F1 score $F1(i)$, a statistical indicator defined in eq. (8), is used to assess the effectiveness of binary classification systems.

$$F1(i) = \frac{2 \cdot \left(\frac{CM(i, i)}{\sum_{m=1}^n CM(i, m)} \right) \cdot \left(\frac{CM(i, i)}{\sum_{m=1}^n CM(m, i)} \right)}{\left(\frac{CM(i, i)}{\sum_{m=1}^n CM(i, m)} \right) + \left(\frac{CM(i, i)}{\sum_{m=1}^n CM(m, i)} \right)} \quad (8)$$

All these performance measures are used to assess the model's performance comprehensively and objectively.

G. LOSS FUNCTION

The loss function measures the discrepancy between the model's predictions and the desired results. We used binary cross-entropy loss in our studies, which is a specific type of cross-entropy (CE) loss designed for binary classification problems given by eq. (9). For binary classification applications, binary cross-entropy loss is well-suited, understandable, and harmonic with gradient-based optimization techniques. Its computation often involves calculating the average cross-entropy across all instances in the dataset, that is:

$$L = -\frac{1}{S} \left[\sum_{i=1}^S (t_i \log(p_i) + (1 - t_i) \log(1 - p_i)) \right] \quad (9)$$

where L represents the loss, S denotes the number of ECG recordings while t_i is the true value taking a value 0 or 1, and p_i is the Softmax probability for the i^{th} ECG signal.

TABLE 3. Model train and val performance metrics.

Epoch	Train Accuracy	Val Accuracy	Train Loss	Val Loss
1	0.5666	0.5044	0.8293	1.1615
10	0.9330	0.8816	0.2076	0.2937
50	0.9813	0.9518	0.0792	0.1355
115	0.9890	0.9693	0.0288	0.0729
163	0.9963	0.9956	0.0190	0.0206

TABLE 4. Gap between train and validation performance.

Epoch	Gap	
	Train & Val. Accuracy	Val. Loss & Val. Accuracy
1	0.0622	0.0322
10	0.0514	0.0861
50	0.0295	0.0563
115	0.0197	0.0441
163	0.0007	0.0016

IV. RESULTS AND DISCUSSION

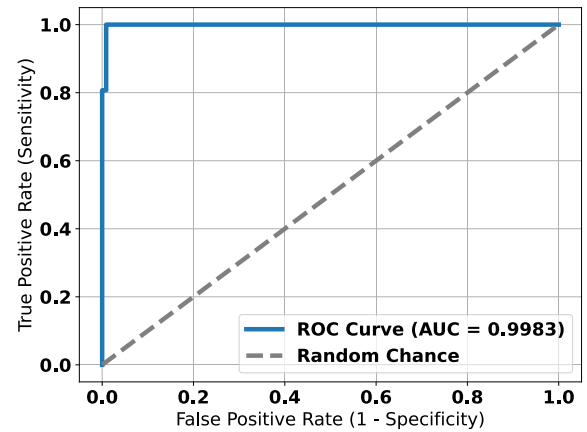
This section provides a detailed evaluation of the proposed model as per different performance metrics. First, the accuracy and loss of the proposed model is evaluated on PhysioNet dataset are provided in Table 3. It can be observed that the model achieved a high validation accuracy of 0.9956 and a low validation loss of 0.0206, indicating that it learned to recognize the ECG patterns effectively. Moreover, from Table 4 that model has a gap between training and validation accuracy of 7×10^{-3} and a gap between training and validation loss of 16×10^{-2} , which are on significantly lower side suggesting that the model is well generalized and learnt the underlying features avoiding overfitting.

Then, the model's performance is evaluated using other performance metrics like Receiver Operating Characteristics (ROC)-Area Under the Curve (AUC), Precision-Recall (PR) curves-AUC, per-class metrics like Positive Predictive Value (PPV, Precision), Sensitivity (Recall, True Positive Rate), macro and weighted averages, F1 score, confusion matrix, and accuracy are calculated.

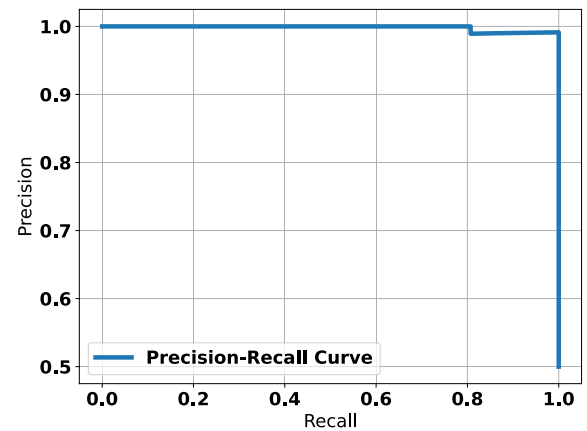
Figure 8 represents ROC-AUC. It measures the model's ability to segregate classes across varying thresholds. On the x-axis is the False Positive Rate (FPR) and on the y-axis is the True Positive Rate (TPR). The ROC-AUC ranges from ROC-AUC $\in [0, 1]$ where ROC-AUC = 1 represents the highest possible ROC-AUC performance. It can be observed that the proposed model has ROC-AUC ≈ 1 .

Figure 9 represents the PR-curve AUC. It measures the positive class performance of the model. On the x-axis is the Recall and on the Y-axis is the Precision. Similar to ROC-AUC, it also ranges from $[0, 1]$ where PR-curve AUC = 1 represents the best performance. It can be observed that the proposed model has PR-curve AUC ≈ 1 .

Table 5 represents the per-class performance in terms of precision, recall, and F1 scores. Moreover, accuracy, macro, and weighted averages are also represented as well. It can be observed that the model exhibits superior performance in terms of the per-class precision, recall, specificity and F1

**FIGURE 8.** ROC-AUC of the proposed hybrid CNN-LSTM model.**TABLE 5.** ECG classification report for test dataset.

Class	Precision	Recall	F1-Score	Support
Normal	1.00	0.9912	0.9956	114
AF	0.9913	1.00	0.9956	114
Accuracy			0.9956	228
Macro Avg	0.9956	0.9956	0.9956	228
Weighted Avg	0.9956	0.9956	0.9956	228

**FIGURE 9.** PR curve-AUC of the proposed hybrid CNN-LSTM model.

scores. It can be observed that the macro and per-class metrics are also equal, reflecting that the model is well-trained and exhibits no bias towards any class. The model also achieved an accuracy = 99.56% and specificity = 99.12%.

Figure 10 represents the performance comparison of the proposed model as compared to the existing models in terms of F1 scores vs. model size. It is to be noted that the model size is a function of learnable and non-learnable parameters. Moreover, the higher the number of parameters, the higher will be the computational cost and will require more time to evaluate an ECG signal. In this regard, a model with the least number of parameters without any compromise on the performance is the most desirable model. It can be observed that the proposed model has the least number of parameters

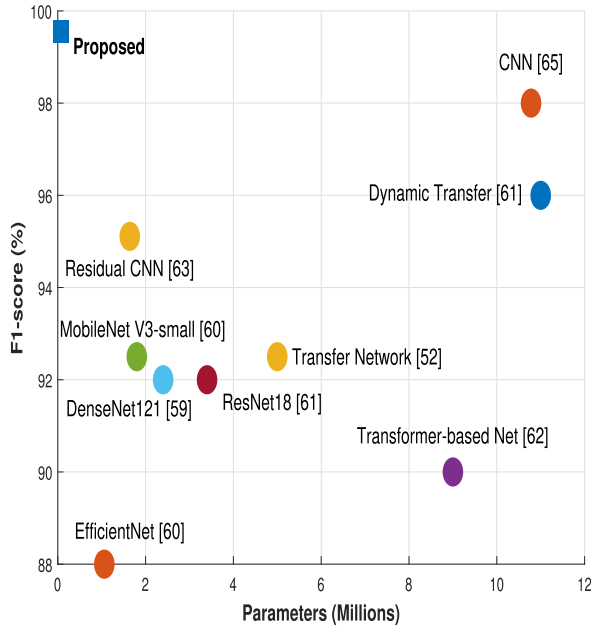


FIGURE 10. Performance characteristics of the proposed model vs. the existing models in terms of F1 score and model size.

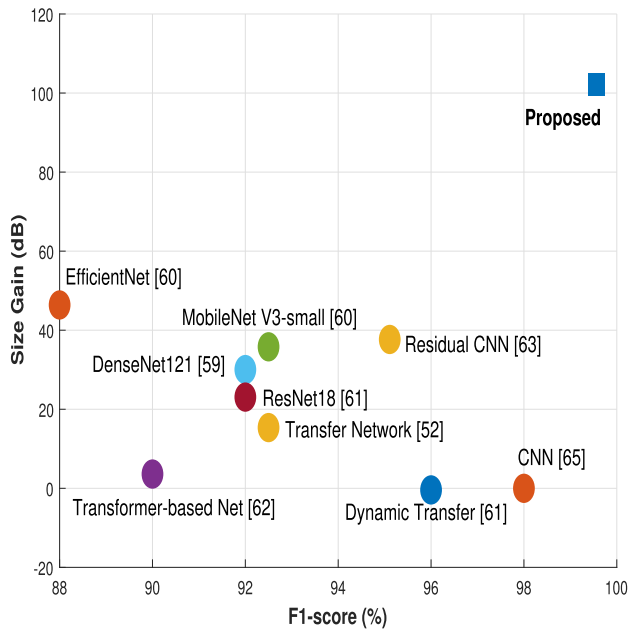


FIGURE 11. Performance characteristics of the proposed model with the existing models in terms of achieved size gain.

and the highest F1 score making it the most suitable model for the ECG classification task.

Figure 11 represents the performance comparison of the proposed model as compared to the other SOTA models in terms of gain achieved due to the smaller model size and F1 score. Specifically, the gain is calculated as

$$Gain_i = 20 \log \frac{Params_{baseline}}{Params_i} \quad (10)$$

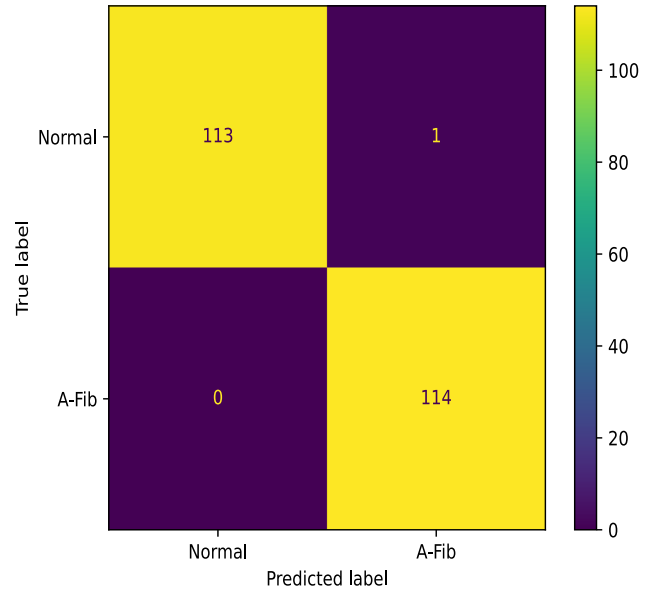


FIGURE 12. Confusion matrix of the proposed model.

where $Params_{baseline}$ represents the parameters of the existing model with the highest F1 score, which in this scenario is CNN based model [65]. The i takes the values from M where $M = \{ \text{Proposed, EfficientNet [60], Transfer Network [52], Transformer-based Net [62], MobileNet V3-small [60], DenseNet121 [59], ResNet18 [61], Dynamic Transfer [61], CNN [65], Residual CNN [63]} \}$ represent the set of models and the parameters in millions are represented by $Params = \{0.064913, 1.06, 5, 9, 1.8, 2.4, 3.4, 11, 10.78, 1.64\}$. The proposed model provides a gain of 102.25 dB while providing the best performance proving its superiority for the ECG classification task as compared to the existing SOTA.

Figure 12 represents the confusion matrix of the proposed model evaluated on the test data. In confusion matrix, the diagonal values need to be as high as possible, and the off-diagonal values need to be as low as possible in order to have a better performance. It can be observed that the proposed model fulfills this criteria with minimum off-diagonal values and diagonal values almost equal to the size of the support. This proves that the model has extracted the features efficiently and generalizes well across the data.

Table 6 shows the comparison of proposed model with the SOTA approaches on the PhysioNet dataset in terms of architecture, input data, data imbalance, and F1-score. The proposed model achieved an F1-score of 99.56%, providing a performance gain of 1.06% without using any preprocessing steps.

Finally, we evaluated the execution time of the proposed model to evaluate its performance for the real-time classification of the ECG signals. To measure the inference time to classify the ECG signal by the proposed model, we evaluated a total of 228 signals, with 114 signals from each class. We averaged it over the total number of signals. The inference

TABLE 6. Comparison of the proposed model with state-of-the-art approaches.

Year	Architecture	Input Data	Address Data Imbalance	F1 Score
2018	13-layer 1D CNN [58]	Segment repetitions or zero padding up to 18,286 samples	No	0.83
2021	Mobile Net V3 [60]	Raw signal	No	0.94
2022	Efficient Net [61]	Raw signal	Yes	0.88
2022	Residual Network [62]	Multiscale encoding of electrocardiogram signals	No	0.98
2022	Transformer-based CNN-LSTM model [63]	Raw signal	No	0.90
2023	DenseNet121, MobileNetV2, ResNet18 [64]	Raw signal	No	0.92
2023	Deep Spectral Features [64]	Raw/1D ECG signals to Mel spectrogram	No	0.90
2024	Transfer Network [65]	Raw signal	No	0.92
2024	CNN [66]	Preprocessed ECG signal	No	0.98
2024	Residual CNN [67]	Compressed ECG signal	No	0.95
2024	MuDANet (Baseline Paper) [67]	9S ECG signal	No	0.96
2025	Our Model	Raw signal	Yes	0.9956

s

TABLE 7. ECG classification report for test dataset (without augmentation).

Class	Precision	Recall	F1-Score	Support
Normal	0.9645	0.9265	0.9451	762
AF	0.6111	0.7719	0.6822	114
Accuracy			0.9064	876
Macro Avg	0.7878	0.8492	0.8136	876
Weighted Avg	0.9185	0.9064	0.9109	876

time of the proposed model is 0.4843×10^{-3} sec making it possible to analyze ECG signals in real-time.

V. ABLATION STUDIES

In this section, we provide the details of the various settings to achieve the performance specified by Table 3. Specifically, modifications are performed in terms of 1) size of augmented data 2) model hyperparameters and performance metrics are observed for the validation set.

A. DATA AUGMENTATION

Table 7 represents the performance of the proposed model without data augmentation. It can be observed that without data augmentation, the model lacks reliable performance across the classes and cannot handle class imbalance. To resolve class imbalance, we performed training on the proposed model for 3 different-sized datasets generated as per the Algorithm 1. The details of each experiment is provided in Table 8. It can be observed that by increasing the sample

TABLE 8. Model performance under variable sized dataset.

No. of Samples/class	F1-score
3000	97.04 %
5040	99.56 %
7000	99.25 %

TABLE 9. Model performance under variable learning rate.

Learning Rate	F1-score
0.1	98.25 %
0.01	98.67 %
0.001	99.56 %
0.0001	99.20 %

size from 3000 to 5040 increased; however, a marginally decreased performance is observed by further increasing the sample size, indicating overfitting.

B. MODEL HYPERPARAMETERS

For hyperparameter tuning, various settings are simulated while keeping the dataset fixed with the best F1 score. Under these settings, the performance is observed by varying the learning rate and dropout values for the LSTM and dense layers. The impact of these variations is given as follows:

Dropout: The dropout is used for the LSTM and dense layers. It was observed that 0.2 dropout for LSTM and 0.5 dropout for fully connected layers provided the best performance. A lower dropout for LSTM explains that the relationship between sequential information is more important for ECG classification.

Learning Rate: A range of learning rate values was applied, the impact of which is provided in Table 9. It can be observed that the best generalization is achieved when the learning rate is set 0.001 and otherwise the model starts to memorize some of the parameters.

VI. CONCLUSION

In this paper, we introduce an ultra-lightweight architecture for binary classification of sinus and atrial fibrillation (A-Fib) rhythms, directly analyzing raw ECG signals to skip lengthy preprocessing and boost information retention. To address the imbalanced PhysioNet ECG dataset, we applied three unique filters—Neurokit, BioSPPy, and EngZeeMod—to generate a balanced dataset. Our model achieved a 99.56% F1 score, with the ultralight design (64.913K parameters) yielding a 102.25 dB size gain and a 1.06% performance improvement despite its small size. With an inference time of 0.48×10^{-3} seconds, it delivers real-time ECG classification, demonstrating its efficiency and reliability for single-lead ECG analysis.

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